

Student handout

Supporting Information for

Title:

Discovery-based approach to identify multiple factors that affect the spin state of coordination complexes using the Evans NMR method

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Use the table to summarize the data

Complex	Concentration of complex	Charge on metal	# of d-electrons	NMR peak shift of TMS	# of unpaired electrons determined by TMS	NMR peak shift of CHCl ₃	# of unpaired electrons determined by CHCl ₃	d-orbital splitting diagram	Spin state
Cr(acac) ₃									
Mn(acac) ₃									
Fe(acac) ₃									
Co(acac) ₃									
Ru(acac) ₃									

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Use the table to summarize the data

Complex	Concentration of complex	Charge on metal	# of d-electrons	NMR peak shift of DSSs	# of unpaired electrons determined by DSS	NMR peak shift of H ₂ O	# of unpaired electrons determined by H ₂ O	d-orbital splitting diagram	Spin state
[Fe(H ₂ O) ₆] ³⁺									
[Co(H ₂ O) ₆] ²⁺									
[Co(H ₂ O) ₆] ³⁺									
Fe(CN) ₆ ⁴⁻									
Fe(CN) ₆ ³⁻									
Co(CN) ₆ ³⁻									